

# SREE GNANESH DOMMETI

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## Education

### University of Maryland, College Park

*M.S. in Applied Machine Learning; GPA: 3.88 / 4.0*

May 2026

College Park, MD

### National Institute of Technology, Calicut

*B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10*

May 2022

Calicut, India

## Experience

### FROOT Lab, University of Maryland

November 2025 – Present

*Research Assistant, Advisor: Alan Zaoring Liu*

College Park, MD

- Contributing to ProjectASAP, an approximation-first telemetry research effort that serves approximate analytics from streaming sketches instead of repeated raw time-series scans.
- Implemented PromQL workload-analysis and sketch-planning paths for high-cardinality aggregations and quantiles, targeting configurable error bounds with transparent fallback to Prometheus.
- Additional NetArena work: built a Dockerized MALT purple agent with LiteLLM prompting, no-helper safety constraints, output validation, retry logic, and local benchmark submission workflows.

### GAMMA Lab, University of Maryland

January 2026 – Present

*Research Assistant, Advisor: Dinesh Manocha*

College Park, MD

- **TraceGround: Grounded Reasoning Evaluation for MLLMs** – building evaluation tooling that scores intermediate reasoning steps against visual/textual evidence.

### Qualcomm

July 2022 – July 2024

*Software Engineer, LTE Modem Software*

Hyderabad, India

- Maintained and debugged LTE Layer 3 RRC modules including Cell Selection, Paging, Security, Handover, and System Information for 4G/5G, CAT-M1, and NB-IoT modem stacks.
- Built Python log-analysis, regression-tracking, and changelist-routing automation over modem traces and CR history, reducing manual triage and engineering assignment effort by 40%.
- Designed fuzz-testing strategies for malformed LTE/NR OTA messages and fake base-station behavior, improving resilience of modem protocol modules against spoofed network signals.

### Connycat

June 2025 – August 2025

*ML Engineer Intern, CampusAI Event Recommendation*

College Park, MD

- Built a personalized campus event recommendation system with FastAPI, Elasticsearch lexical retrieval, MiniLM semantic search, and reciprocal-rank fusion; achieved <300 ms p95 retrieval and <600 ms p95 end-to-end latency.
- Implemented ALS and BPR collaborative filtering over implicit student interactions, improving long-tail event recall; deployed services on AWS EC2 with Docker and MLflow experiment tracking.

## Projects

### CAFBrian - Knowledge Retrieval and Content Generation | *LangChain, LangGraph, FAISS, OpenAI API*

- Built a multimodal RAG system over 10,000+ pages of PDFs, PPTs, DOCX, transcripts, and CAFB communication assets using custom parsing, OCR/layout extraction, chunking, embeddings, and FAISS retrieval.
- Implemented human-in-the-loop LangGraph review flows for grant proposals, blogs, slides, and social drafts; optimized retrieval to <1.2s/query and reduced repetitive content creation time by more than 50%.

### Benchmarking LLM Inference Frameworks | *Python, PyTorch, vLLM, SGLang, Docker, AWS EC2 GPU*

- Deployed vLLM and SGLang to serve Mistral-7B-Instruct-AWQ on EC2 GPU; built a Poisson-arrival benchmarking harness across 1–64 concurrent users for latency, throughput, cost, and GPU utilization.
- Measured SGLang at 4,001 tokens/sec saturation and vLLM with 34% better throughput at 16 users; identified vLLM as more cost-efficient under randomized traffic at \$0.020 per 1M tokens.

### Real-Time Fraud Detection Pipeline | *PyTorch, AWS ECS/Fargate, EvidentlyAI, DVC, ClearML, GitHub Actions*

- Deployed a containerized fraud model on AWS ECS/Fargate with autoscaling configuration, service health checks, and resource tuning for production-style inference.
- Integrated EvidentlyAI drift checks, GitHub Actions CI/CD, DVC data versioning, and ClearML experiment tracking to make model updates reproducible and observable.

## Skills

**Languages:** Python, SQL, C, C++, Rust

**ML / GenAI:** PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace, XGBoost, LightGBM, LangChain, LangGraph, LlamaIndex, RAG, LoRA, DPO, vLLM, SGLang

**Data / Infra:** Kafka, Arroyo, Spark, Redis, PostgreSQL, Elasticsearch, FAISS, Pinecone, Prometheus, Grafana, OpenTelemetry

**MLOps:** Docker, Kubernetes, AWS ECS/EC2/S3/SageMaker/Lambda, Terraform, MLflow, ClearML, DVC, W&B, GitHub Actions, EvidentlyAI